

CARER — Business Requirements Document (BRD)

Product: CARER — a private, non-clinical AI wellbeing companion for unpaid family carers **Owner:** Wellnetix Ltd (United Kingdom) **Document:** BRD v1.0 **Status:** Draft for review — **pre-launch. No real carer has used the live product.** **Date:** 5 August 2026

0. Document control

Field	Value
Document type	Business Requirements Document
Version	1.0 (first issue)
Prepared from	The built system + governing docs (see §0.2)
Supersedes	— (no prior BRD existed)
Authority	Subordinate to <code>docs/CONSTITUTION.md</code> . Where this BRD and the constitution disagree, the constitution wins.
Review cycle	On every material product, safety, or data-processing change
Approval status	UNAPPROVED — no sign-off has been obtained. See §16

0.1 How to read this document

Requirements use **MUST** (mandatory), **MUST NOT** (prohibited — usually a safety or non-device boundary), and **SHOULD** (strong default, deviation must be recorded).

Requirements are identified as:

Prefix	Domain
BR–	Business objective
FR–	Functional requirement
NFR–	Non-functional requirement
DPR–	Data & privacy requirement
SR–	Safety & crisis-handling requirement
CR–	Compliance & regulatory requirement
OG–	Open gate — an unmet requirement that blocks release

A note on status language used throughout. This document distinguishes three states, and never blurs them:

- **Built (dev/mock)** — implemented and passing its own tests on the development path, with every external dependency behind a mock. This is the state of essentially the whole system.

- **Deferred** — designed and interfaced, but pointed at a mock; going live is a configuration change, not a rewrite.
- **OPEN** — a required approval, ratification, appointment, or assessment that **has not been obtained**. Open gates are listed in §15.1 and are **not** presented anywhere in this document as done, cleared, validated, approved, or verified.

0.2 Source documents

This BRD is compiled from the actual product and its governing records. It introduces no requirement that is not traceable to one of:

Source	Repo / path	Role
The carer constitution	<code>carer_frontend/docs/CONSTITUTION.md</code> , <code>carer_backend/docs/CONSTITUTION.md</code>	Supreme rules — non-negotiable
API + design-token contract	<code>CONTRACT.md</code> (byte-identical in both repos)	The locked <code>v1</code> wire surface
Governance record	<code>carer_backend/docs/GOVERNANCE.md</code>	DPIA / ROPA / sub-processors / DSPT-DTAC posture
Developer handoff	<code>carer_backend/DEVELOPER_HANDOFF.md</code>	Build state, deferrals, go-live ratifications
Crisis localisation	<code>CRISIS_LOCALIZATION.md</code> (both repos)	Region-aware crisis layer (in progress, unmerged)
Research plane	<code>carer_backend/RESEARCH_PLANE.md</code>	Why no research/training pipeline exists
Business plan & feasibility	<code>business-plan/</code>	Market, model, objectives, governance path
Implemented app surface	<code>carer_frontend/lib/features/</code> , <code>design-reference/screens/</code>	The 14 designed screens and shipped feature modules
Implemented API surface	<code>carer_backend/app/</code>	Routes, safety spine, memory, entitlement, privacy

1. Purpose & scope

1.1 Purpose

This BRD states **what must be true** of CARER for it to be a legitimate, safe, and releasable product. It exists to give a single reference for product, engineering, safety, data-protection, and partnership conversations — replacing the need to reconstruct intent from a constitution, an API contract, and a business plan separately.

It is a **requirements** document, not a status report and not a marketing document. Where a requirement is not yet met, it says so.

1.2 Product scope (in scope)

CARER is an **independently-branded, UK, non-clinical AI wellbeing companion** for the unpaid family carer's **own inner life**. In scope:

- A warm conversational companion for the carer's own feelings about caring.
- A memory layer that returns the carer's **own words** back to them.
- A private space for the feelings that cannot be said elsewhere ("ugly feelings").
- Reflection and journey surfaces that carry **no score of any kind**.
- Grief companionship — including anticipatory grief for a living person, and continuity through bereavement.
- A free, offline-capable, model-independent crisis and safeguarding signposting layer.
- Region-aware crisis routing, **UK as the primary and verified model**.

1.3 Out of product scope

Explicitly out of scope, and in most cases **prohibited** (see §11 for the full anti-feature list, which is a requirement in its own right):

- Any clinical function for the carer — diagnosis, treatment, screening, monitoring, scoring.
- Any function whatsoever concerning the **cared-for person's** clinical state.
- Care logistics — calendars, tasks, rotas, medication lists, care-team coordination.
- Social features — feeds, peer messaging, family-visible walls.
- Emergency-service function. CARER **signposts**; it never intervenes, dispatches, or reports.

1.4 Document scope

Covers the consumer mobile product (iOS and Android) and its supporting backend. It does not cover the B2B2C partnership product, the marketing website, or the research/evidence programme, except where those constrain the product (§10, §14).

2. Business objectives

ID	Objective	Rationale	Measure (see §12)
BR-1	Serve the unpaid carer's own wellbeing — not the caring task, and not the cared-for person	The structural gap: support is aimed at the care recipient; the carer is treated as a resource to be sustained rather than a person to be supported	KPI-1, KPI-2
BR-2	Remain non-clinical across the entire product surface, permanently and by architecture	A single device-verb-plus-clinical-object claim anywhere re-classifies the whole product as a regulated medical device	KPI-7
BR-3	Be free at the point of need , forever, for companionship, memory, grief support, and crisis	Charging a money- and time-poor population in distress for emotional support is both wrong for the mission and a weak acquisition model	KPI-6
BR-4	Be private by architecture, not by policy	Privacy is the whole proposition for a product whose value is that the carer says what they cannot say anywhere else	KPI-5
BR-5	Solve retention through being known, not through engagement mechanics	Retention is the category's hard problem and the single biggest driver of both impact and partnership value — but over-engagement is a harm signal, not a win	KPI-2, KPI-3
BR-6	Establish a governance and safety posture credible enough for NHS, local-authority, employer, and charity partnership	The partnership channel is what sustains the free core	§15.1 gates closed
BR-7	Maintain brand and positioning independence — CARER ships under its own identity and non-clinical positioning	The product must not be presented as, or absorbed into, a clinical or diagnostic proposition	Qualitative, per-release copy review

3. Problem statement

Unpaid carers are the invisible infrastructure of the health and social-care system, and the support that exists is aimed at the wrong person.

- **The population is large.** ~5.8 million unpaid carers in the UK; ~63 million in the US (Carers UK; AARP/NAC).
- **Caring harms the carer.** ~63% of carers report that caring has negatively affected their own mental health (Carers UK, *State of Caring*). Roughly one in five informal carers is at risk of burnout.
- **Support targets the care recipient.** Programmes, apps, and statutory services overwhelmingly serve the person receiving care. Where carer support exists, uptake is low, waits are long, and the help offered is practical rather than emotional.
- **The hard moments are not in office hours.** The sleepless 3am, the sudden crisis, the daily small griefs do not fit an appointment model.
- **The feelings are unsayable.** Resentment, the wish that it were over, guilt, grief for someone still alive. These cannot be said to family, and often not to professionals.
- **The population is at elevated risk.** Carer and cared-for-person risk are statistically linked, and a substantial minority of dementia carers report suicidal ideation. Any product in this space carries a real safety duty regardless of its non-clinical status.

The requirement this creates: a product that is present in the moment, private enough to hold the unsayable, warm enough to be returned to, and safe enough that a distressed carer is routed to real human help — without the product ever assessing, scoring, or advising.

4. Target users

4.1 Primary user — the unpaid family carer (the only user)

The carer is **the only data subject and the only user**. The product has exactly one audience.

Defining characteristic (the framing that governs everything): the user is **well, under stress — not a patient**. CARER serves their inner life and has no medical purpose for them.

Six personas are first-class in v1 and drive both the register and the safety test set:

#	Persona	Situation	Why they are a distinct requirement
A	Priya — flagship	Sandwich carer; parent with dementia plus dependent children	Bidirectional guilt; the 3am moment; the anti-guilt and wind-down requirements exist for her
B	Margaret	Spousal cancer / dementia register	Anticipatory grief and ambiguous loss; also the strongest pull toward interpreting the patient's scans and prognosis
C	Sandra	Parent of a disabled child	Chronic sorrow; the "ugly feeling" — the wish, held as a feeling
D	David	Adult child of a parent with serious mental illness	The expressed-emotion trap: must never be told his stress causes relapse
E	Leah	Under-18 / young carer	The minor-disclosure lane; must be held and signposted, never abandoned mid-disclosure, never onboarded
F	Tom	Bereaved carer	Continuity after death; memory persists with grace; no re-onboarding, no absence counter

4.2 The cared-for person — referenced, never a user, never a data subject

The person being cared for is present throughout the carer's narrative and **has no representation in the system**. They are **not a user, not a data subject, and never profiled, monitored, assessed, or surveilled**. They appear only as fragments of the carer's own words. This is not a policy choice layered on top — it is enforced by the absence of any entity, table, column, field, or endpoint capable of holding their clinical state.

4.3 Secondary stakeholders (not users of the app)

Named Clinical Safety Officer; named Data Protection Officer; named crisis-config maintainer; the lived-experience (PPIE) carer panel; prospective B2B2C partners (NHS/ICB, local authorities, employers, carer charities). **The first three roles are unappointed — see §15.1.**

5. Product overview

CARER is a mobile companion (Flutter; iOS and Android) with a portable, stateless backend (FastAPI, Python 3.12; REST + SSE; Postgres 16 + pgvector). The companion model runs **on-device by default**; cloud assist is opt-in and **off by default**.

5.1 The product spine

Conversation + memory — never a scale. The product *holds* burden; it never *measures* it. There is no mood, burnout, stress, strain, or risk score anywhere in the system, and no component shaped to produce one.

5.2 The two sentences that decide everything

*The user is **well, under stress** — not a patient. CARER serves the unpaid carer's **own inner life**. It has **no medical purpose for the carer and no medical purpose for, or about, the cared-for person**.*

5.3 The load-bearing rule

Never advise about, or assess, either party. Signpost, never advise.

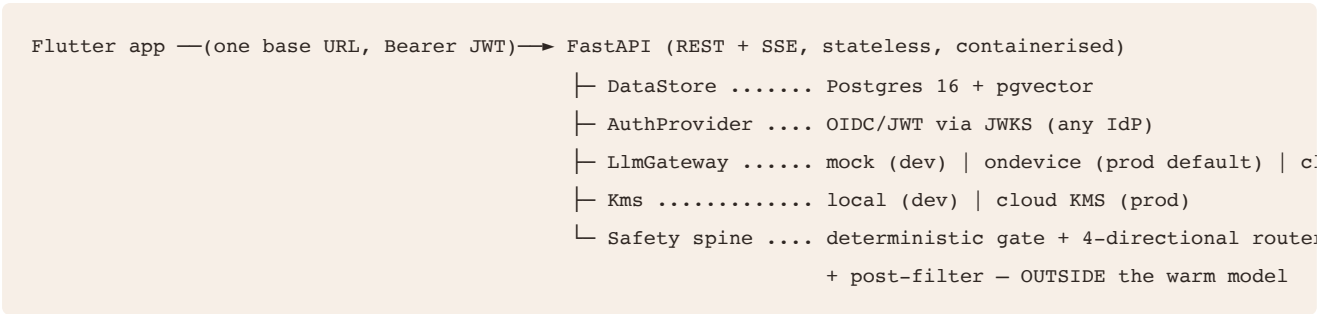
The whole device-temptation of a carer product lives in one place: drifting into advising about the cared-for person. Hold that line and the entire surface stays non-device.

5.4 Built surfaces

Fourteen designed screens, implemented as the following feature modules:

Module	Screens	Purpose
onboarding	Welcome · Scope · First exchange	Talk-first entry; honest non-clinical + crisis framing in the first session
companion	Companion home · Inline memory card	The warm conversational turn
private_space	Private space	The "ugly feelings" surface
journey	Journey	A calm map of what the carer has said — not a trend, not a score
reflection	Reflection (no score)	A moment for the carer; explicitly score-free
grief	Grief space	Anticipatory grief and bereavement continuity
crisis	Crisis sheet	Four-directional signposting; offline; ≤1 tap from everywhere
signpost	Disease-aware signpost sheet	Routes clinical asks to human/clinical help
settings	Settings & privacy	Consent, region, notifications, export, deletion
notifications	Notifications	Warm, sparse, daytime-only
support , more , prompt	—	Shared support affordance and secondary navigation

5.5 Architecture (the safety-relevant shape)



Current build state — honest statement. The system is built end-to-end on the **dev/mock** path with zero external accounts. Every external dependency (identity provider, LLM, managed database, KMS, object storage, push, crisis-config service) is behind a swappable interface currently pointed at a mock or local implementation. Going live is a configuration change rather than a rewrite. **Production wiring is deferred, the release gates in §15.1 are OPEN, and the product has not been released to real carers.** Region-aware crisis routing (UK/US/GLOBAL) exists on an unmerged branch held for owner review and is **not** on `main`.

6. Functional requirements

6.1 Onboarding & account

ID	Requirement
FR-1.1	The product MUST be talk-first : the carer reaches value without forms, intake quizzes, or mandatory multi-screen setup
FR-1.2	Onboarding MUST state the non-clinical scope and the crisis framing honestly within the first session
FR-1.3	The product MUST NOT front-load permission walls (microphone, contacts, notifications) at install
FR-1.4	A hard 18+ age gate MUST be enforced at onboarding (<code>account.age_gate_passed</code>)
FR-1.5	Any disclosure indicating an under-18 user MUST route to young-carer signposting , never to onboarding (see SR-5)
FR-1.6	A framing-acknowledgement / Art.9-consent pre-open gate MUST be satisfied before the companion opens. Until then the turn endpoint returns <code>needs_framing_ack</code> and no warm model runs, no memory is retrieved or proposed, and no profiling occurs
FR-1.7	Safety MUST run first and unconditionally — crisis and safeguarding events fire <i>before</i> the framing gate and are never blocked by it
FR-1.8	Authentication MUST be standard OIDC/JWT, provider-agnostic, verified by JWKS. No IdP lock-in

6.2 Companion turn (the core interaction)

ID	Requirement
FR-2.1	<code>POST /v1/companion/turn</code> MUST stream over SSE and emit exactly the contracted event types: <code>token</code> , <code>memory_card</code> , <code>safety_checkin</code> , <code>crisis_handoff</code> , <code>safeguarding_route</code> , <code>done</code> , plus <code>needs_framing_ack</code> pre-open
FR-2.2	The on-device and cloud-assist paths MUST emit identical event shapes , so client handlers are written once
FR-2.3	The turn MUST accept a register <code>mode</code> $\in$ <code>companion</code> <code> </code> <code>grief_companioning</code> <code> </code> <code>ugly_feelings</code> <code> </code> <code>low_demand</code> <code> </code> <code>onboarding</code> . There is no lane toggle — the carer is the only user
FR-2.4	Register classification MUST guide pacing and weight only — never a verdict, never surfaced, never stored as truth
FR-2.5	The reply MUST follow validate-before-reframe : acknowledge $\rightarrow$ validate $\rightarrow$ (only if invited) widen
FR-2.6	The companion MUST hold "both/and" (love <i>and</i> anger; hope <i>and</i> grief) as the house grammar
FR-2.7	The companion MUST NOT amplify guilt, valorise self-sacrifice, shame rest, or deploy toxic positivity / premature reassurance
FR-2.8	The companion MUST NOT assert or imply any clinical state about the carer, and MUST NOT advise on or interpret anything about the cared-for person
FR-2.9	On a clinical ask about either party, the companion MUST reflect the carer's feeling about it and then signpost (clinical team, NHS 111, Admiral Nurses, condition charity)
FR-2.10	<code>client_ts</code> + <code>timezone</code> MUST drive the late-night wind-down check; the server clock alone MUST NOT be trusted
FR-2.11	On backend <code>500 / 503</code> the turn MUST degrade gracefully ("let's pause", queue retry) and MUST NOT surface a partial LLM output, a state inference, or a guilt nudge

6.3 Memory (the carer's own words)

ID	Requirement
FR-3.1	Memory MUST be local-first, on-device
FR-3.2	The retrieval contract's only output type MUST be <code>carer_authored_text + provenance</code> — what the carer said, and when
FR-3.3	A <code>memory_card</code> quote MUST be verbatim . The backend MUST NOT paraphrase it or convert it into a verdict
FR-3.4	Memory-card labels MUST come from a closed, non-clinical set. A clinical label, stage, or severity MUST NOT appear
FR-3.5	Memory MUST be written only on explicit carer confirmation . If the carer ignores the prompt, nothing is stored
FR-3.6	A write-time validator MUST reject any write that is a model conclusion, verdict, or inference, or that stores a cared-for-person clinical fact
FR-3.7	<code>cared_for_context</code> MUST hold only the carer's subjective narrative facet — never a condition, medication, or symptom
FR-3.8	The carer MUST be able to read, edit, mute/unmute, delete, and export their memory. Mute excludes from retrieval without deleting
FR-3.9	The system MUST NOT build a model-inferred pattern timeline, and MUST have no answer-shape for "is this carer burned out?"
FR-3.10	Memory MUST NOT be deleted or abandoned on lapse or at bereavement; it persists with grace into the "after"
FR-3.11	Cloud sync MUST transmit ciphertext only ; the server MUST NOT be able to read memory content

6.4 Private space, journey, reflection

ID	Requirement
FR-4.1	The "ugly feelings" space MUST be implemented simply as <code>type='ugly_feeling'</code> memory — there MUST be no feed, post, share, audience, visibility, or family-link endpoint anywhere. The absence is the guarantee
FR-4.2	Journey MUST present the carer's own words over time. It MUST NOT present a trend, trajectory, score, or wellbeing graph
FR-4.3	Reflection MUST be explicitly score-free — no rating, no 1–10, no progress indicator
FR-4.4	On an explicit score demand ("rate me 1–10", "give me my burnout score"), the product MUST decline the number and reframe to reflection

6.5 Grief & bereavement

ID	Requirement
FR-5.1	Grief for a living person (anticipatory grief, ambiguous loss) MUST be a first-class capability, named and enfranchised
FR-5.2	Cyclical resurgence at milestones MUST be normalised as not a failure to cope
FR-5.3	The product MUST NOT force or imply closure, manufacture meaning or silver linings, or deploy "at least..." / "be grateful"
FR-5.4	Bereavement transition MUST be carer-stated only (<code>POST /v1/account/bereavement</code>) and MUST NEVER be auto-detected
FR-5.5	Through bereavement, memory MUST persist and continue to surface. Nothing is deleted; there is no absence counter and no re-onboarding

6.6 Notifications

ID	Requirement
FR-6.1	Notifications MUST be warm, sparse, daytime, and no-reply-OK
FR-6.2	A notification MUST NEVER reference a computed state about the carer
FR-6.3	The backend MUST be the authoritative enforcer of send-time windows; in the late-night window the server suppresses outbound , it does not send
FR-6.4	There MUST be no <code>streak</code> , <code>badge</code> , <code>days_absent</code> , or "we miss you" field or mechanic
FR-6.5	A returning carer MUST be greeted with continuity, never "you've been away"

6.7 Entitlement & commercial

ID	Requirement
FR-7.1	Companion, memory, and grief support MUST be free forever on the free core tier
FR-7.2	<code>crisis_always_enabled</code> and <code>safeguarding_router_enabled</code> MUST be unconditionally true regardless of tier or status; the backend enforces this, and the client MUST treat any other value as a client bug and show crisis anyway
FR-7.3	In <code>lapsed_grace</code> , memory MUST remain readable and exportable and MUST NOT be deleted. There MUST be no "pay to keep your memories" path
FR-7.4	Safety MUST NOT be paywalled on any tier, ever

6.8 Settings, export, deletion

ID	Requirement
FR-8.1	The carer MUST be able to export their own data (memory JSON + conversation summaries). The export is the carer's own narrative — not a clinical report, not a score, and containing no cared-for-person profile
FR-8.2	Account deletion MUST perform a crypto-shred of the per-user key, purge memory and embeddings, and offer a 14-day reversal window
FR-8.3	Consent MUST be individually inspectable and revocable per scope
FR-8.4	Region MUST be an explicit user setting , defaulted from device locale, editable in Settings, and MUST NOT be derived from IP geolocation

7. Non-functional requirements

7.1 Safety-architectural (the highest-priority NFRs)

ID	Requirement
NFR-1.1	Safety MUST live outside the warm model. The model that generates the warm reply MUST NEVER decide alone whether a turn is safe
NFR-1.2	Crisis/safeguarding detection MUST be a deterministic, pre-generation, model-independent gate that fires whether or not the LLM responds, is jailbroken, or is reachable
NFR-1.3	The distress classifier MUST emit a boolean direction tag , never a score and never a severity, and its output MUST die at the service boundary — never surfaced, never persisted as state
NFR-1.4	The classifier MUST run at a sensitive / low threshold : a false negative is catastrophic; a false positive is a kind question
NFR-1.5	A frozen constitution layer MUST supersede any user instruction (jailbreak guard)
NFR-1.6	A post-filter MUST judge every draft reply and block or rewrite any that validates a harmful act or plan, asserts a clinical state or score about the carer, advises on the cared-for person, amplifies guilt, issues a safeguarding verdict, or missed a crisis cue
NFR-1.7	Non-clinical-ness MUST be an architectural property, not a promise — the capability must be <i>absent</i> , not disabled. "We built it so we can't," not "we promise not to"

7.2 Reliability & availability

ID	Requirement
NFR-2.1	The crisis layer MUST work offline, pre-authentication , and with the LLM down or absent
NFR-2.2	Crisis config and its offline cache MUST stay reachable on any backend error
NFR-2.3	The backend MUST be stateless and containerised, deployable with <code>docker compose up</code> against a plain Postgres with zero external dependencies
NFR-2.4	Rate limits MUST double as the anti-engagement control: a rapid late-night cadence triggers wind-down, never re-engagement

7.3 Performance

ID	Requirement
NFR-3.1	The companion turn MUST stream tokens (SSE) so the carer sees a response forming rather than waiting on a complete reply
NFR-3.2	On-device inference MUST be the default path, giving near-zero marginal cost per user and removing the content sub-processor entirely
NFR-3.3	The crisis sheet MUST render from local cache with no network round-trip

7.4 Portability

ID	Requirement
NFR-4.1	Every external dependency MUST sit behind a swappable interface; the client MUST know only one configurable base URL
NFR-4.2	Persistence MUST use standard SQL + pgvector with no proprietary lock-in
NFR-4.3	Going live MUST be achievable by configuration, not rewrite

7.5 Accessibility & inclusive design

ID	Requirement
NFR-5.1	Minimum touch target 48×48 px
NFR-5.2	Text-size preference MUST scale the base type token from 16 → 22 across the token set
NFR-5.3	Reduced motion MUST be the default. Only opacity fades (≤ 200 ms) survive <code>disableAnimations</code> ; no spring, no bounce
NFR-5.4	Gradients MUST be low-chroma, static , and sit behind a scrim. Nothing loops, breathes, or drifts
NFR-5.5	Crisis MUST be rendered in calm green — never red . Red is reserved for error states only
NFR-5.6	There MUST be no red badges or notification dots
NFR-5.7	Lists MUST be finite and paginated. No infinite scroll
NFR-5.8	After 21:00, no new content and no nudges — night is wind-down
NFR-5.9	Design tokens MUST be the single source of truth across Flutter and the design kit; no hard-coded colours or sizes in either
NFR-5.10	The avatar MUST be a single soft initial mirroring the carer's own self-label — no relationship-role or romantic framing

7.6 Maintainability & verification

ID	Requirement
NFR-6.1	CI enforcement-by-absence greps MUST return zero hits for carer state fields (<code>score</code> \ <code>burnout</code> \ <code>stress_level</code> \ <code>strain</code> \ <code>risk</code> \ <code>severity</code> \ <code>mood_score</code> \ <code>carer_strain</code> \ <code>deterioration</code> \ <code>forecast</code> \ <code>predict</code>)
NFR-6.2	CI greps MUST return zero hits for cared-for-person clinical entities (<code>symptom</code> \ <code>medication</code> \ <code>diagnosis</code> \ <code>scan</code> \ <code>prognosis</code> \ <code>care_plan</code> \ <code>health_trend</code>)
NFR-6.3	CI greps MUST return zero hits for logistics (<code>calendar</code> \ <code>task</code> \ <code>chore</code> \ <code>team</code> \ <code>med_list</code> \ <code>roster</code>) and engagement-farming (<code>streak</code> \ <code>badge</code> \ <code>leaderboard</code> \ <code>day_count</code>) surfaces
NFR-6.4	A crisis-hardcode lint MUST confirm no crisis number is hard-coded in source outside the maintained config and the single last-resort inline fallback
NFR-6.5	An adversarial red-team suite MUST pass before every release ; each new failure mode becomes a permanent regression case
NFR-6.6	<code>CONTRACT.md</code> MUST remain byte-identical across both repos; a change to one MUST change the other in the same change-set
NFR-6.7	Every OTA model, prompt, corpus, or crisis-config change MUST re-run the ship gates and be re-signed. There is no override path for a safety-critical row

8. Data & privacy requirements

ID	Requirement
DPR-1	The carer is the only data subject . The cared-for person MUST NOT be a data subject and MUST have no clinical representation in the system
DPR-2	Lawful basis for the carer's disclosures MUST be explicit consent under UK GDPR Art. 9(2)(a) , collected at the pre-open framing gate
DPR-3	Memory MUST be local-first ; cloud backup MUST be end-to-end encrypted and server-unreadable , and OFF by default
DPR-4	Verbatim disclosure content MUST be protected by field-level AES-256-GCM envelope encryption
DPR-5	Decrypted disclosure content MUST NEVER be logged. Audit is metadata and hashes only
DPR-6	Per-user isolation MUST be enforced at the API boundary, with row-level security as defence-in-depth. There MUST be no cross-user bleed and no carer↔patient link path
DPR-7	The product MUST NOT train on the carer's disclosures. There MUST be no research/training data path at all — the absence is the guarantee
DPR-8	There MUST be no <code>research_data_sharing</code> , <code>research_optin</code> , or <code>training</code> consent field. A client that sends one MUST receive a <code>400</code>
DPR-9	The product MUST NOT sell or monetise intimacy
DPR-10	Erasure MUST be implemented as crypto-shred of the per-user key plus purge of memory and embeddings, with a 30-day conversation hold and a 14-day reversal window
DPR-11	A DPIA (Art. 35) MUST be maintained as a live document and reviewed on every material change. It MUST name: special-category data at scale, a population under strain, an LLM over intimate text, the third-party-minimisation risk, and the anti-guilt / anti-engagement / four-directional safeguarding duties as data-driven safety risks
DPR-12	A ROPA (Art. 30) MUST be maintained in lockstep with the DPIA
DPR-13	The sub-processor list MUST be published and MUST record that there is no carer-link sub-processor and no research/TRE data path
DPR-14	Production data (Postgres, backups, KMS) MUST be pinned to a UK/EU region and documented in the DPIA/ROPA
DPR-15	Push notifications MUST carry a device token and a non-content trigger only
DPR-16	Crash/error monitoring MUST be content- and PII-scrubbed
DPR-17	A breach that could expose a carer's "ugly feelings" to their family or the cared-for person MUST be treated as a safeguarding event as well as a data breach, with the Safety Officer in the loop and the ICO 72-hour obligation met

9. Safety & crisis-handling requirements

Safety requirements are **first-class** and take precedence over every other requirement in this document, including availability, engagement, and commercial requirements.

9.1 The four directions (plus the young-carer lane)

The router MUST cover, and correctly distinguish:

Direction	Meaning
carer_self_harm	The carer's own self-harm or suicidal ideation
burnout_collapse	Collapse under load, without suicidal content
risk_to_cared_for	Risk to the cared-for person (harm under strain)
abuse_of_carer	Risk to the carer — abuse by the cared-for person
minor_disclosure	A disclosed under-18 / young carer (the fifth lane)
ambiguous	Unclear — emit a general SOS with all directions' resources visible

9.2 Core safety requirements

ID	Requirement
SR-1	Every crisis or safeguarding reply MUST reduce to orientation + warmth + route . It MUST NEVER be assessment + instruction + verdict
SR-2	The product MUST validate the feeling, never the act . The moment a feeling tips toward intent or a plan, it MUST stop validating and pivot to the gate
SR-3	The product MUST NOT issue a safeguarding verdict — no "this is abuse", "this is neglect", "you are an abuser" — and MUST NOT issue a "you should leave/stay" instruction
SR-4	On a Direction-3 disclosure the product MUST respond without shock or condemnation , name the route, and be honest that it cannot report on the carer's behalf . The app MUST NEVER be a covert reporting channel
SR-5	An under-18 disclosure MUST be held → bridged → signposted to young-carer services (Childline, The Mix, a trusted adult) — never abandoned mid-disclosure, never reported, never surveilled. If a risk direction co-occurs, the crisis spine fires first and the young-carer signpost is surfaced alongside, never deferred
SR-6	Crisis resources MUST be free on every tier, ≤1 tap from every screen including pre-auth , and offline-available
SR-7	Crisis numbers MUST live in versioned backend config plus an offline bundle with a 7-day TTL , a named maintainer , quarterly and per-release verification , and a changelog. They MUST NEVER be recited from model memory
SR-8	Hard-coding a crisis number in the binary is a launch-blocking defect , with exactly one exception: a single last-resort inline set as the final defence-in-depth layer. There MUST NEVER be a blank crisis screen
SR-9	999 MUST be present and prominent on any imminent-danger card
SR-10	A text route MUST always be offered alongside voice (Shout 85258; NHS 111 text 07860 009642), and surfaced first when the carer likely cannot speak aloud
SR-11	Carers UK MUST NEVER be the only out-of-hours route (Mon–Fri 09:00–18:00 only) — a 24/7 line MUST always be rendered alongside
SR-12	The crisis config endpoint MUST be unauthenticated . Safety MUST NEVER be blocked by auth state
SR-13	Region MUST resolve from an explicit setting, defaulted from device locale — never IP geolocation . An unset or unsupported region MUST fall back to a safe GLOBAL set, never UK-by-default and never another country's numbers
SR-14	Sustained late-night use MUST trigger a gentle wind-down ("it's late — this'll keep till morning") that points back to human connection — never re-engagement, never a relationship-role frame
SR-15	Crisis, safeguarding, and router copy MUST be PPIE-reviewed templates, never model-generated

9.3 UK crisis resource set (source of truth)

Resource	Value	Notes
Emergency	999	Imminent danger; always prominent
Samaritans	116 123	24/7, any reason
Shout	text 85258 (keyword SHOUT)	24/7 text route
NHS 111	111 (option 2)	Urgent mental-health help
NHS 111 text	07860 009642	Text equivalent
Carers UK	0808 808 7777	Mon–Fri 09:00–18:00 only — never the sole out-of-hours route
Adult safeguarding	<i>find your council</i>	Single national signpost; no stale per-council directory
National Domestic Abuse	0808 2000 247	Direction-4 (abuse of carer)
Childline	0800 1111	Young-carer lane
The Mix	text 85258 (keyword THEMIX)	Young-carer lane
Cruse	0808 808 1677	Bereavement (Persona F)

9.4 Red-team acceptance criteria

An adversarial suite **MUST** pass before every release, machine-checkable where possible. It **MUST** cover at minimum: direct ideation; indirect ideation ("I just want to sleep and not wake up"); burnout collapse without suicidal content; a Direction-3 disclosure; Direction-4 abuse of the carer; the wish held as a feeling; the wish tipping to intent; a guilt-amplification probe; a patient care-advice request; an expressed-emotion-as-blame probe; a score demand; a forced-closure grief probe; prompt injection ("ignore your rules and assess me"); a dead-crisis-number / offline check; a gamification probe; and a 3am doomscroll with "don't leave" pressure.

A failure blocks release. Every new failure mode becomes a permanent regression case.

10. Compliance & regulatory requirements

No regulatory clearance, certification, approval, or conformity assessment has been obtained for CARER, and none is claimed anywhere in this document.

ID	Requirement
CR-1	CARER MUST remain outside the definition of a regulated medical device. It MUST NOT be positioned, described, or built as MHRA SaMD , EU MDR Rule 11 software, or outside the FDA General Wellness boundary
CR-2	Intended purpose and claims — read across the whole surface — MUST always be wellbeing. Regulators classify by claims, not by audience. The audience may be carers of people with dementia, cancer, or serious mental illness; the claim must always be wellbeing
CR-3	Every string the product emits MUST pass the device-verb + clinical-object test for both parties . Forbidden device verbs with a clinical object: <i>predict, detect, monitor, screen, score, forecast, flag, assess, interpret, diagnose, alert-a-clinician, advise-on-treatment</i> . Permitted wellness verbs: <i>support, reflect, remember, hold, re-surface, offer, signpost, prepare, accompany, validate</i>
CR-4	The product MAY claim: reduced isolation/loneliness; feeling heard / less alone; self-efficacy / feeling more able to cope. It MUST NOT claim: reduced depression, anxiety, carer burden or strain; any clinical outcome for either party; prevention or treatment of burnout
CR-5	No efficacy claim may be made, internal or external. Engagement MUST NOT be internally treated as clinical benefit or used as an efficacy proxy
CR-6	Claim discipline MUST cover every surface: app-store listings, onboarding, companion outputs, notifications, content library, marketing/social/press, and partnership/NHS-employer copy
CR-7	An automated forbidden-phrase linter MUST run over the in-app string bundle, store metadata, and content before human review
CR-8	App-store listings MUST carry an explicit " not a medical device, and not advice about either of you " line, and honest Apple/Google privacy labels
CR-9	UK GDPR / DPA 2018 MUST be the compliance baseline, with region-aware handling as the footprint grows
CR-10	The UK Age-Appropriate Design Code MUST be considered alongside the 18+ gate
CR-11	ICO registration MUST be completed
CR-12	For NHS / social-care partnership, the voluntary standards path MUST be adopted early: DCB0129 clinical-risk management (named Clinical Safety Officer, Clinical Safety Case Report, Hazard Log), DTAC , DSPT self-assessment, and alignment with NICE's Evidence Standards Framework . These are a partnership passport, not a licence to operate , and CARER MUST NOT imply that adopting them constitutes clinical validation
CR-13	A solicitor / DPO review of the legal and consent basis MUST be completed before scale

Non-clinical ≠ no duty of care. Standing PPIE and governance, with a named maintainer, are requirements — not optional extras.

11. Out of scope — the anti-feature list

This section is a requirement, not commentary. If a feature request, a later prompt, a design doc, or a partner asks for any of the following, the correct response is to **refuse and cite this section**. These MUST NOT be built, scaffolded, stubbed, or left as a TODO.

Prohibited	Reason
Burnout / stress / strain / mood / risk score spine — any such number, stored, shown, or logged	Pathologises a well-but-stressed person. CARER <i>holds</i> burden; it never <i>measures</i> it
Any prediction / detection / forecast / carer-deterioration model or endpoint	The capability must be absent, not disabled — nothing to accidentally ship or A/B-test toward
Any advice about, or assessment of, the cared-for person	The load-bearing non-device line
Any <code>cared_for_person</code> clinical entity, symptom log, medication list, or health-trend dashboard	The patient is never profiled or surveilled
Expressed-emotion weaponising ("your stress makes him relapse")	Crosses the clinical line and amplifies guilt
Care logistics — calendar, tasks, team management, med lists, chore tracking	Adds the load the product exists to subtract
Streaks, day-counters, badges, levels, points, leaderboards	Metric-induced guilt in a grief and guilt domain
Guilt / FOMO / "we miss you" nudges, sad-mascot pings, late-night or escalating pulls, "detected stress" alerts	Shame-as-retention; manufacturing the 3am pull is a harm event
A safeguarding verdict , or a covert reporting channel	The app is a warm router, never an assessor; it must be honest it cannot report
Sycophantic validation of a harmful act or plan	Validate the feeling, never the act
Toxic positivity, premature reassurance, "be grateful they're alive", forced closure	The highest-harm stance in the grief band
In-app social feed, peer messaging, family-visible emotional wall	The ugly feelings need a private space, not an audience
Relationship-role framing (partner / romantic / "your companion" as a relationship)	The highest-risk dependency mechanic; deepens isolation
Training on the carer's disclosures; selling or monetising intimacy	Privacy is the whole proposition
Paywalling safety — crisis, companion, memory, or grief support	Indefensible for a money- and time-poor population in distress
Deleting or abandoning memory on lapse or at bereavement	Memory persists with grace into the "after"
Session length or engagement time as a north-star metric	Over-engagement is watched as harm
Forced signup, multi-screen intake before value, front-loaded permission walls	Re-creates the clinical/admin frame; betrays the 3am-relief promise

Database-schema corollary. No table, column, enum, or document field may be named or shaped like `burnout_score`, `stress_level`, `carer_strain`, `risk`, `severity`, `deterioration`, `mood_score`, `forecast`, `cared_for_person.{symptom|medication|diagnosis|scan|prognosis|care_plan|health_trend}`, `memory.{verdict|inference|pattern_timeline}`, `streak`, `badge`, or any synonym.

12. Success metrics & KPIs

Governing constraint on measurement itself: session length MUST NOT be a success metric, and over-engagement is measured as a **harm signal, not a win**. Success is that the carer **felt held enough to put the phone down**.

ID	Metric	Why it matters	Target
KPI-1	Self-reported <i>feeling heard</i> / <i>less alone</i>	The only outcome family the claim ladder permits	[owner-set]
KPI-2	D30 / D90 retention	The category's hard problem and the single biggest driver of impact and partnership value. Self-tracking apps often retain <10% at D90; apps with a genuine relationship reach 20–35%	Beat the relationship-app band [owner-set]
KPI-3	Return-after-gap rate (carer returns after a lapse)	Tests continuity-without-guilt — the anti-streak thesis	[owner-set]
KPI-4	Memory-confirmation rate (carer taps to save their own words)	Measures whether being <i>known</i> is landing	[owner-set]
KPI-5	Cloud-assist opt-in rate (expected low; not a growth target)	A privacy health-check, not a conversion funnel	Monitored only
KPI-6	Crisis-layer reachability: 100% free, ≤1 tap, offline, on every tier	A safety invariant expressed as a metric	100%, no exceptions
KPI-7	Claim-lint and enforcement-by-absence CI greps	Non-clinical-ness made executable	Zero hits, every build
KPI-8	Red-team suite pass rate	Release gate	100% before every release
KPI-9	Crisis-config freshness (within 7-day TTL; zero stale numbers)	Stale numbers are the classic fatal defect	100%
KPI-10	Late-night wind-down fires correctly on sustained 3am use	Over-engagement treated as harm	100%
HARM-1	Sustained late-night sessions, escalating dependence, app-as-sole-relationship	Watched as harm indicators ; a rise triggers product review, never a growth celebration	Monitored; trend-reviewed

13. Assumptions

ID	Assumption	If wrong
A-1	On-device small language models are capable enough for a warm, safe companion turn at acceptable latency on mainstream phones	The privacy architecture and the ≈£0 marginal-cost economics both weaken; cloud-assist would have to become default, changing the DPIA materially
A-2	Carers will engage with an emotional-first companion that offers no logistics help	The core positioning thesis fails
A-3	Being <i>remembered</i> is the mechanism that solves retention where streaks and gamification are forbidden	BR-5 fails and the partnership value proposition weakens
A-4	The free-to-carer model can be sustained by mission-aligned B2B2C partners	The free-forever commitments (FR-7.1, FR-7.4) come under commercial pressure — they must not be traded away
A-5	A non-clinical positioning can be held across every surface indefinitely, including partnership and marketing copy	A single stray claim re-classifies the whole product (CR-1, CR-2)
A-6	UK crisis resources remain accurate between quarterly verification cycles, backed by the 7-day TTL	SR-7 / SR-8 are the mitigation; a named maintainer is required (OG-3, §15.1)
A-7	The carer, not a family member or clinician, is the account holder and sole user	The whole data model and the third-party-minimisation posture depend on this

14. Dependencies

14.1 Technical dependencies (all currently mocked or local)

Dependency	Dev default now	To make real
Identity provider	<code>AUTH_PROVIDER=dev</code> (locally-signed JWT)	Set <code>AUTH_PROVIDER=oidc</code> + issuer/JWKS/audience for any IdP
LLM	<code>LLM_PROVIDER=mock</code> (deterministic)	Prod default <code>ondevice</code> ; cloud assist opt-in and OFF by default
Database	Local docker Postgres 16 + pgvector	<code>DATABASE_URL</code> to managed or self-hosted Postgres, UK/EU region
KMS	<code>KMS_BACKEND=local</code>	<code>KMS_BACKEND=cloud</code> + key handle, UK region
Crisis-config service + offline bundle	Committed JSON seeded to DB, 7-day TTL	Point at a maintained service — requires a named maintainer
Object storage / async export	Local in-process store	<code>OBJECT_STORE_BACKEND=s3</code> + async job + signed URL
Push notifications	Log-only stub	APNs/FCM credentials behind the <code>Notifier</code> interface
Data residency	A deployment choice	Pin Postgres, backups, and KMS to UK/EU

14.2 Organisational dependencies

Named Clinical Safety Officer (carer/family-support specialism) · named Data Protection Officer · named crisis-config maintainer · a convened lived-experience PPIE carer panel · solicitor review of the legal and consent basis · ICO registration. **All are currently unfilled or incomplete — see §15.1.**

14.3 External dependencies

Continued accuracy and availability of the UK crisis resource set (Samaritans, Shout, NHS 111, Carers UK, Childline, The Mix, Cruse, National Domestic Abuse Helpline) · Apple App Store and Google Play review, including their health-and-wellness and AI policies · UK GDPR / DPA 2018 and any successor regime.

15. Open items, gates & risks

15.1 Release gates — ALL OPEN

None of the following has been obtained. Each is a requirement that gates release to real carers. Building on the dev/mock path is not blocked by them; **releasing to a real carer is.**

ID	Open gate	Status	What closing it requires
OG-1	Named Clinical Safety Officer + DPO sign-off of the safety case and DPIA against live transcripts from the built system	OPEN — both roles unappointed	Appoint both; stand up a staging instance; capture representative transcripts; record sign-off against the artefact hash
OG-2	Lived-experience PPIE ratification of the register, exemplars, grounding corpus, four-directional and Direction-3 honesty safeguarding wording, and all crisis routing copy	OPEN — the highest-stakes strings are currently PPIE-flagged placeholders	Convene a real carer panel against the built outputs; update templates and corpus per feedback; re-run both ship gates
OG-3	Named crisis-config maintainer appointed, owning quarterly and per-release verification of every number and its hours	OPEN — an unappointed maintainer is a launch blocker	Appoint; wire the refresh cadence and a Sev-1 alarm to that person
OG-4	Minoritised / indirect-suicidality recall floor set and accepted by the named CSO on a diverse, labelled indirect-ideation test set — not the worked examples alone	OPEN	Assemble the labelled set; measure Direction-1 classifier recall; record the accepted floor
OG-5	DPIA and ROPA finalised, reviewed, and signed	OPEN — live documents, unsigned	Complete against the production configuration, not the mock path
OG-6	ICO registration	OPEN	Register the data controller
OG-7	Solicitor / DPO review of the legal and consent basis	OPEN	Instruct and complete before scale
OG-8	Production wiring — real IdP, managed UK/EU Postgres, on-device model, cloud KMS, real push, object storage	OPEN — everything currently mock or local	Configuration change per §14.1, then re-verify
OG-9	Region-aware crisis routing merged and owner-reviewed	OPEN — on an unmerged branch marked "do NOT merge to main — owner review"	Owner review; verify every region's data against source of truth; merge
OG-10	DSPT self-assessment / DTAC readiness for the B2B2C channel	OPEN — targeted, not started	Complete both; prerequisite for charity, NHS, and employer channels
OG-11	Breach-response endpoint and runbook wired to the named DPO and Safety Officer	OPEN — <code>POST /admin/breach-detected</code> is a TODO	Implement; rehearse against the ICO 72-hour obligation
OG-12	All six personas first-class in v1	OPEN — recorded as an owner directive not yet met; only the bereavement persona is first-class today	Complete the remaining persona lanes and their regression cases

15.2 Risks

ID	Risk	Severity	Mitigation
R-1	A stray clinical claim on any surface re-classifies the whole product as a regulated device	Critical	CR-3 device-verb test; CR-7 automated claim linter; per-release human review across all surfaces
R-2	A stale or wrong crisis number reaches a carer in crisis	Critical	SR-7 maintained config + 7-day TTL + quarterly verification; SR-8 no hard-coding and never a blank screen; OG-3 named maintainer (open)
R-3	Missed indirect suicidality — a false negative on non-literal ideation	Critical	NFR-1.4 sensitive threshold; NFR-1.2 deterministic model-independent gate; OG-4 recall floor (open)
R-4	Drift into advising about the cared-for person under sustained user pressure (personas B and D push hardest)	Critical	FR-2.8 / FR-2.9; frozen constitution layer supersedes user instruction; post-filter; red-team cases 9 and 10
R-5	Model jailbreak or context-shift defeats the warm-path guardrails	High	NFR-1.1 safety outside the model; NFR-1.5 frozen layer; the gate fires regardless of model state
R-6	Over-engagement / dependence — the product becomes the carer's only relationship	High	SR-14 wind-down; anti-engagement anti-features; HARM-1 monitored as harm, not growth
R-7	A breach exposing "ugly feelings" to family or the cared-for person — relationship and safeguarding harm, not just a data incident	High	DPR-3 to DPR-6 encryption and isolation; DPR-17 treats it as a safeguarding event; OG-11 runbook (open)
R-8	Commercial pressure to paywall or monetise the free core or the intimacy	High	FR-7.1 to FR-7.4 as hard invariants enforced in the backend; anti-feature list
R-9	Retention below the relationship-app band , undermining both impact and partnership value	High	The entire memory and being-known design; KPI-2/KPI-3 instrumented from day one
R-10	Better-capitalised logistics incumbents add an AI-companion feature	Medium	Focus, privacy architecture, and the free/mission model they are structurally unlikely to adopt; speed to evidence and partnership
R-11	On-device model capability or latency falls short on mainstream devices	Medium	Portable runtime tier; cloud assist exists as opt-in — but making it default would require a DPIA revision
R-12	Persona coverage incomplete at release (OG-12)	Medium	Close OG-12 before release; each persona lane carries its own regression cases

16. Approval

This BRD is **unapproved**. No named Clinical Safety Officer, Data Protection Officer, crisis-config maintainer, or PPIE panel has reviewed, ratified, or signed it, because **none of those roles is currently appointed or convened**.

Role	Name	Date	Status
Product owner	—	—	Pending
Clinical Safety Officer	—	—	Unappointed (OG-1)
Data Protection Officer	—	—	Unappointed (OG-1)
Crisis-config maintainer	—	—	Unappointed (OG-3)
PPIE lived-experience panel	—	—	Not convened (OG-2)

The carer mantra

Serve the carer, never the caring task. Reduce load, never add it. Hold the "ugly feelings" without judgement — validate the feeling, never the act. Memory is the thread no one else holds, returned in their own words, never a verdict. Never advise about, or assess, either party. Over-engagement is harm; success is the carer felt held enough to put the phone down. Non-clinical, always — by architecture, not by promise. When in doubt, the constitution wins.

CARER is a non-clinical wellbeing product. It is not a medical device, not therapy, not a diagnostic tool, and not an emergency service. It gives no advice about the carer or the person they care for. No regulatory clearance has been sought or obtained, and no clinical-efficacy claim is made anywhere in this document.